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Prevalence of suicidal behaviours among medical students in South Asia in the 1st quarter of the 21st century: a systematic review and meta-analysis protocol

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ABSTRACT

Background Medical students in South Asia face intense mental and psychosocial challenges, increasing their vulnerability to suicidal behaviours. Despite sporadic studies, no comprehensive meta-analysis has yet examined the prevalence and associated risk factors in this region. Limited mental health resources and underreporting further obscure the true burden. This study protocol aims to systematically assess suicidal behaviours among South Asian medical students.

Methods and analysis This systematic review and meta-analysis will encompass Medline (PubMed), Scopus, Web of Science, EMBASE, PsycINFO, Google Scholar and regional databases for studies published between 1st January 2000 and 1st May 2025 on suicidal behaviours, adhering to Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols and Meta-Analysis of Observational Studies in Epidemiology guidelines. Cross-sectional, longitudinal and cohort studies using validated assessment tools and including South Asian medical students, published without any language restriction will be included. Excluded studies will include review articles, intervention studies, commentaries, preprints, meeting abstracts, protocols, unpublished research and letters. Screening, data extraction and quality assessment will be conducted independently by reviewers using tools such as EndNote and Rayyan, with quality assessed via the modified Newcastle-Ottawa Scale. Data synthesis will involve calculating pooled prevalence estimates using a random effect meta-analysis models, where applicable, narrative synthesis of risk factors. Forest plots will visualise pooled estimates, heterogeneity will be assessed using the I^2 statistic, and funnel plots along with Egger's test will be used to evaluate publication bias. Subgroup and sensitivity analyses will be conducted if substantial heterogeneity is detected. Statistical analysis will be conducted using R(v.4.3.2) with built-in meta-packages and GraphPad Prism v.9.

Ethics and dissemination As this review involves only pre-published data, ethical approval is not required. The region-specific findings will be submitted to an international peer-reviewed journal and shared through scientific conferences, webinars and relevant platforms to disseminate the burden of suicidal behaviours among South Asian medical students. Key insights will be disseminated to policymakers, educators, clinicians and researchers to inform suicide prevention strategies and guide future research.

PROSPERO registration number CRD420251032718.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ The present study will rigorously focus on estimating the pooled prevalence of suicidal behaviours among medical students in South Asia.
⇒ The review will adhere strictly to the Cochrane handbook, Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols, Meta-Analysis of Observational Studies in Epidemiology and modified Newcastle-Ottawa Scale guidelines and corresponding checklists at each stage.
⇒ Only peer-reviewed studies, regardless of language, that report primary data on South Asian medical students will be included.
⇒ Potential heterogeneity in study designs and assessment tools may limit generalisability of meta-analysis and pooled estimates.
⇒ A limited number of studies and reliance on self-reported measures may introduce methodological and reporting bias.

INTRODUCTIONS

The medical education environment both in medical schools and universities serves as a critical setting for the academic, social, political and professional development of students, particularly in the demanding field of medical science. This period is marked by significant academic pressure, life transitions, personal hardship and the need to make important personal and professional decisions.[1 2] For medical students, this phase often involves adapting to intense workloads, forming new social networks and navigating

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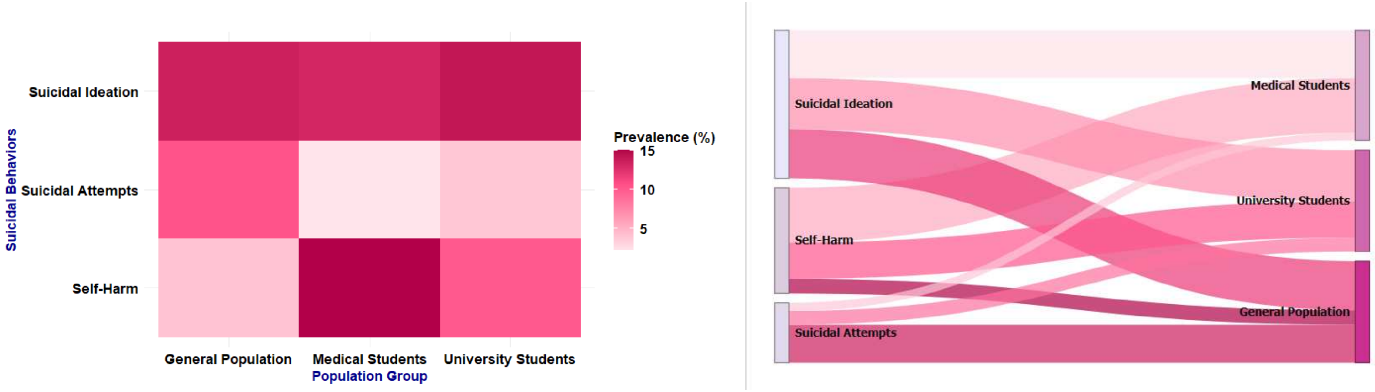


Figure 1 Prevalence of global suicidal behaviours among different population group. The heatmap and sankey diagrm illustrating the prevalence of suicidal behaviours among medical students, university students and the general population was generated by author (MR) using R v.4.3.2 based on aggregated data from the cited studies. References: Medical Students[1 2]; University Students;[2 3] General Population.[9 10] Data Source: Mudiyanselage *et al*[9]; Crispim *et al*[3]; Zimmermann *et al*[10]; Seo *et al*, 2021[17] and Blacker *et al*.[22]

high expectations.[1] While these experiences may offer opportunities for personal growth, diversity and resilience. The demanding structure and pressures of medical college training, combined with limited systemic support system may, contribute to psychological distress, potentially increasing the risk of suicide among medical students.[3]

Suicidal behaviour is a multifaceted phenomenon encompassing suicidal ideation (SI), suicidal attempts (SA) and suicidal planning (SP), influenced by a range of biological, psychological, social and environmental factors.[3–5] Suicide attempts often represent the culmination of a gradually evolving process. Understanding the context in which suicidal behaviour emerges is essential for identifying and mitigating its contributing factors.[6] According to the WHO reports that over 720 000 people die by suicide each year worldwide, with a global age-standardised rate of 9.2 per 100 000 population. The highest rates are observed in Europe (12.4 per 100 000), followed by South-East Asia (10.1 per 100 000). Suicide was identified as the third-leading cause of death among individuals aged 15–29 years while 73% occurred in low-income and middle-income countries (LMICs).[7] However, the rate of suicide among the medical students ranged from 3.8% to 18.7% which is notably 3–5 times higher compared with general population[5 8–10] (figure 1).

South Asia, comprising eight countries[11] (Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka), is predominantly made up of LMICs (~Gross Domestic Product (GDP) per capita USD $2385),[12] where mental health resources are often limited despite a high burden of mental health issues and suicide rates.[13] A systematic review and meta-analysis of 46 studies among South East Asian populations reported that the prevalence of SI (17.4%), SP (9%), SA (5.2%),[14] while the global prevalence of SA (2.19%) was reported in another review.[8] Ahmed *et al*[15] reviewed depression and suicidal behaviours in the WHO, South-East Asia regions over the last 60 years (1956–2016), identifying only 19 studies, while Rahman *et al*[5] reported just only 6 studies on suicidal behaviours among medical students in Bangladesh over the past 24 years (2000–2024), highlighting the limited and uneven research coverage across the region. Furthermore, a scoping review reported suicide rates in South Asia ranged from 0.43 to 331 per 100 000, likely underestimating the true figures due to variations in study populations, methodologies and data registration systems, still higher comparing to world average.[16]

Regarding the factors contributing to the higher suicide risk among medical students globally include poor mental health, such as depression (OR: 6.87), burnout (OR: 6.29), comorbid mental illness (OR: 5.08) and stress (OR: 3.72). Additionally, smoking (OR: 1.92), familial history of mental illness (OR: 1.72) and alcohol use (OR: 1.41) were also identified as risk factors associated with various suicidal behaviours among medical students globally.[17] Most South Asian countries are classified as LMICs, and evidence suggests that the prevalence of SA among medical students is significantly higher in LMICs compared with those in high-income countries (4.02% vs 1.60%).[8] Individual (childhood adversity, bullying, symptoms suggestive of attention-deficit/hyperactivity disorder, unpleasant emotions, anhedonia, sexual orientation, personality disorders, drug abuse) and environmental factors (violence, socioeconomic inequality, lifestyle, lack of social support, media effect and cultural stigma) may further contribute to greater vulnerability to suicidal behaviours in this region. These complexities underscore the need to understand suicidal behaviour from a multifactorial perspective.[3 18–20]

The true extent of suicidal behaviours among medical students in South Asia may be underestimated due to limited evidence[14] and significant underreporting—one of the major challenges in suicidology—hindering accurate estimation of suicide rates in this vulnerable group.[21 22] Although several sporadic studies have examined suicidal behaviour among medical students in South Asia, no comprehensive meta-analysis has been conducted

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to date. Moreover, the potential contributing factors have not been collectively analysed. A systematic exploration focusing on both the prevalence and associated factors of suicidal behaviours in this population remains notably lacking. Addressing this gap is essential to comprehensively understand the burden of suicidal behaviours and to inform evidence-based strategies. Therefore, this protocol aims to systematically synthesise available data to estimate the prevalence and identify key risk factors of suicidal behaviours among medical students in South Asia.

Objective and research question

The primary objective of this study is to systematically review the most recent literature up to April 2025 and conduct a meta-analysis to estimate the pooled prevalence of suicidal behaviours among medical students in South Asia. Additionally, the study aims to find the associated factors and differentiate between fatal and non-fatal SAs, to categorise the prevalence based on SI, SAs and SP.

The research question for this study is guided by the PECO framework,[23] which provides a structured approach by outlining the Population (P), Exposure (E), Comparison (C) and Outcome (O). The skeleton outlines a clear and comprehensive overview. In this context, the Population (P) includes South Asian medical students at the undergraduate, postgraduate and internship levels. The Exposure (E) refers to their enrolment in medical schools and the Outcome (O) focused on suicidal behaviours. As this is a prevalence study, no Comparison (C) group is included.

Guided by PECO framework, this systematic review and meta-analysis will be guided by the research question: 'What is the prevalence of suicidal behaviours among medical students in South Asia?'. Estimating the pooled prevalence of suicidal behaviours including both fatal and non-fatal SAs across eight South Asian countries will allow for regional comparisons and contribute to the broader field of suicide research and prevention. Most importantly, the findings will underscore the need for a comprehensive, multimodal approach to suicide prevention that addresses a wide range of contributory factors beyond mental health morbidity alone.

METHODS

Study design

This systematic review and meta-analysis will adhere to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P 2015)[24] and the Meta-Analysis of Observational Studies in Epidemiology (MOOSE)[25] guidelines. The review will include published studies between 1st January 2000 and 1st May 2025. The study protocol has been registered in PROSPERO[26] under the registration number CRD420251032718.

Eligibility criteria

The inclusion and exclusion criteria are outlined as follows in table 1.

Information sources

The data for this systematic review and meta-analysis will be sourced from a comprehensive range of reputable databases, including Medline (PubMed), Scopus, Web of Science, EMBASE, PsycINFO, Google Scholar and other regional databases (e.g., BangalJOL, NepJOL). Additionally, to ensure the review standard, the reference lists of included studies and relevant systematic reviews will also be thoroughly examined for any additional eligible publications.

Search strategy

Preliminary searches will be conducted using Medical Subject Headings (MeSH) in PubMed to identify relevant keywords associated with the research topic across mentioned electronic and other regional databases accordingly. To ensure a comprehensive and systematic approach, a detailed search strategy will be developed in consultation with an experienced in systematic reviews (librarian). The strategy will combine both MeSH terms and free-text keywords related to the two core concepts of "Suicidal Behaviours" and "South Asian medical students". A preliminary version of the PubMed-specific search strategy is presented in table 2.

Boolean operators (AND, OR, NOT), truncations and database-specific filters will be used to optimise and narrow the search results. To ensure comprehensive coverage, all searches will be performed without applying any restrictions on publication language. Moreover, studies published between 2000 and 2025 will be included to ensure the review captures contemporary research from the 1st quarter of the 21st century that reflects the most robust, recent evidence on suicidal behaviours among South Asian medical students.

Condition/domain being studied

The conditions being studied include suicidal behaviours and associated factors among medical students in South Asia.

Population/participants

The review will encompass studies from diverse ethnic groups and genders across South Asia, covering those pursuing degrees such as Bachelor of Medicine and Surgery (MBBS), Bachelor of Dental Surgery (BDS) from the medical schools considering medical graduates, as well as undergraduate medical students (from first to fifth year of medical curriculum), intern doctors, pre-clinical and clinical students, and both resident and non-resident medical graduate trainees.

Exposure

Being medical students.

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Table 1 Inclusion and exclusion criteria for the systematic review on suicidal behaviours among South Asian medical students

Criteria type	Details
Inclusion criteria	
Study design	Empirical observational studies with quantitative estimates (cross-sectional, case–control, longitudinal including cohort studies) published before 1 May 2025.
Population	Medical students including both MBBS/BDS (undergraduate, postgraduate, internship level) from the eight South Asian countries.
Outcome	Studies reporting overall or separate prevalence of suicidal behaviours (SI, SA, SP) and/or associated factors.
Assessment tools	Use of standardised and psychometrically validated tools (e.g., SBQ-R, PHQ-9, GSHS, BDI, CSSR-S, DSM, ICD-10).
Language and year	Studies published in any language from 1st January 2000 to 1st May 2025
Exclusion criteria (Studies)	
Publication type	Review articles, case reports, case series, media reports, interventional studies, study protocols, commentaries, book chapters, preprints, conference abstracts, letters without primary data.
Data type	Articles with qualitative outcomes only.
Specific subgroup focus	Studies focusing only on specific subgroups (eg, medical students with chronic illnesses).
Population data	Studies not focused on or not specifying the exact number of medical students.
Outcome reporting	Studies lacking quantifiable prevalence data on suicidal behaviours or details on associated factors.
Access	Unpublished studies or studies with inaccessible full texts.
Assessment tools	Studies not using validated tools to assess suicidal behaviours.
Geographic specificity	Articles presenting data from multiple countries without a South Asia-specific subgroup analysis.
Duplicate reporting	Multiple articles published from the same project/data.
Exclusion criteria (population)	
Non-target population	Non-medical students.
Geographic scope	Studies on medical students outside South Asia.
Mixed populations	Studies on healthcare or university students without a subgroup analysis specific to medical students.

BDI, Beck Depression Inventory; CSSR-S, Columbia-Suicide Severity Rating Scale; DSM, Diagnostic and Statistical Manual of Mental Disorders; GSHS, Global School-based Student Health Survey; ICD-10, International Classification of Diseases, 10th Revision; MBBS/BDS, Bachelor of Medicine and Surgery/Bachelor of Dental Surgery; PHQ-9, Patient Health Questionnaire-9; SA, suicidal attempt; SBQ-R, Suicidal Behaviors Questionnaire-Revised; SI, suicidal ideation; SP, suicidal planning.

Comparator(s)/control

There will be no comparison group.

Context

To explore the prevalence of suicidal behaviours (SI, SP and SA) and associated factors among medical students in South Asian region.

Outcome

The primary outcome of this review protocol would be to identify the prevalence of suicidal behaviours (SI, SP and SA) and the secondary outcome would be to explore the factors associated with suicidal behaviours among medical students in the South Asian region.

Study selection process

To ensure a comprehensive and unbiased inclusion of studies relevant to suicidal behaviours among medical students in South Asia, the study selection will follow a multistage screening process. Initially, two independent reviewers will screen the titles and abstracts of all identified records using the predefined inclusion and exclusion criteria. Any discrepancies at this stage will be reviewed and resolved by a third reviewer to ensure consistency and accuracy. Studies deemed potentially eligible will proceed to full-text screening. The full texts will be independently and separately reviewed by two reviewers to determine final eligibility. In addition, inter-rater agreement between reviewers will be assessed at each stage of the review process using Cohen's kappa statistic and percentage agreement. A kappa value ≥ 0.60 will be considered acceptable, indicating substantial agreement.[27 28] Any discrepancies will be resolved through discussion or adjudication by a third reviewer to ensure methodological rigour and consistency. The entire selection process will be transparently documented using the PRISMA flow diagram,[24] detailing the number of records

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Table 2 Search strategy for PubMed database (PICO framework)

Search element	Search terms and structure
Condition (C)	(Suicidal behaviour OR Suicidal ideation OR Suicidal planning OR Suicide OR Suicidal intention OR Suicidal thoughts OR Self-harm OR Suicidal risk OR Suicidality OR Attempted suicide OR Deliberate self-harm)
Population (P)	(Medical students OR Dental students OR Intern doctors OR Undergraduate medical students OR Postgraduate medical students OR Residents OR Residency students) AND (South Asia OR Afghanistan OR Bangladesh OR Bhutan OR India OR Sri Lanka OR Maldives OR Nepal OR Pakistan OR Lanka OR Ceylon)
Outcome (O)	(Epidemiology OR Prevalence OR Trends OR Associated factors OR Determinants OR Contributing factors OR Risk factors OR Protective factors OR Aggravating factors OR Predictors)

PICO: **Patient/Population/Problem (P), Intervention (I), Comparison (C), Outcome (O)**.

identified, screened, excluded and finally included in the review. This systematic approach will help to ensure that only studies of adequate quality and direct relevance to the research question and study outcome are included in the final synthesis.

Reference management and screening platform

To manage the records retrieved through the comprehensive literature search, EndNote v.21.0 (Clarivate Analytics, Philadelphia, USA)[29] will be employed for organising citations. This software will be used to compile all identified references, and duplicate entries will be systematically removed. Any additional studies found through manual searches will also be incorporated into the EndNote library as needed. Following deduplication, the remaining articles will be exported to the Rayyan QCRI,[30] a web-based tool designed to facilitate collaborative screening. Titles and abstracts will be reviewed on the Rayyan platform by independent reviewers during the initial screening phase, with full-text articles uploaded for subsequent assessment. This approach will streamline the selection process and enhance reviewer coordination. A visual summary of this process will be presented in the PRISMA flow chart (figure 2).

Data extraction and management

Following selection of eligible studies, data extraction will be independently conducted by two reviewers using a predesigned, structured extraction form. This form will capture essential study characteristics, including author name, year of publication, study design, sample size, study location, sample size, sampling technique and participants' demographics (eg, age, gender, year of study, academic curriculum (MBBS/BDS), suicidal behaviour). The primary focus will be on extracting data related to the prevalence of suicidal behaviours (SI, SP and SA) among the South Asian medical students. Information on the validated assessment tools such as Patient Health Questionnaire-9 (PHQ-9) will be used to evaluate suicidal behaviours and reported severity levels will also be recorded. For each, the effect measure will be extracted as prevalence proportion (% and f [frequency of outcome] / N [sample size]) along with 95% confidence intervals (CIs). In addition to prevalence data, the reviewers will extract information on the factors associated with suicidal behaviours and methods used for suicidal behaviours. The secondary outcome data relating to risk factors and associated variables will be extracted as ORs or risk ratios with 95% CIs (adjusted or unadjusted) and corresponding p values. Validated assessment tools used to assess suicidal behaviour and extract reported cut-off scores, classification criteria and severity levels (e.g., mild/moderate/severe SB) will also be extracted. Where available, we will also extract qualitative or quantitative data on the methods used in SAs, as reported in the included studies. Furthermore, all the extracted data will be entered into a Microsoft Excel spreadsheet. A third reviewer will verify the accuracy of the data entry to ensure consistency and minimise errors. Any discrepancies will be resolved through consensus or, if needed, with the involvement of a fourth reviewer. The final extracted dataset will be checked and analysed using R v.4.3.2 statistical software to facilitate a comprehensive and transparent synthesis of the findings.

Statistical analysis

To ensure a comprehensive and accurate synthesis of findings, multiple steps will be employed in the analysis of the data. Initially, the prevalence of suicidal behaviours, including SI, SA and SP will be calculated for each study included in the review. In addition, the pooled prevalence will also be calculated separately for fatal and non-fatal suicide attempts. The pooled prevalence estimate will then be computed using a random-effects model to account for the heterogeneity across the studies.[31]

The inverse variance method will be used to calculate the weighted average, allowing for the integration of results from studies with varying sizes and variabilities.[32–34] For studies reporting ORs or relative risk, these measures will be converted to a common metric—namely the log odds ratio (ln (OR))—to facilitate interpretation and consistency across studies.[35–37]

Meta-analysis will be performed using R statistical software (v.4.3.2) with the 'meta' package.[38] To quantify heterogeneity across studies, the I^2 statistic will be reported, with values exceeding 50% indicating substantial heterogeneity.[39] Subgroups will be stratified by characteristics such as type of suicidal behaviours (SI, SA and SP), degree level (eg, undergraduates vs postgraduates, medical (MBBS) vs dental (BDS)), biological sex (eg,

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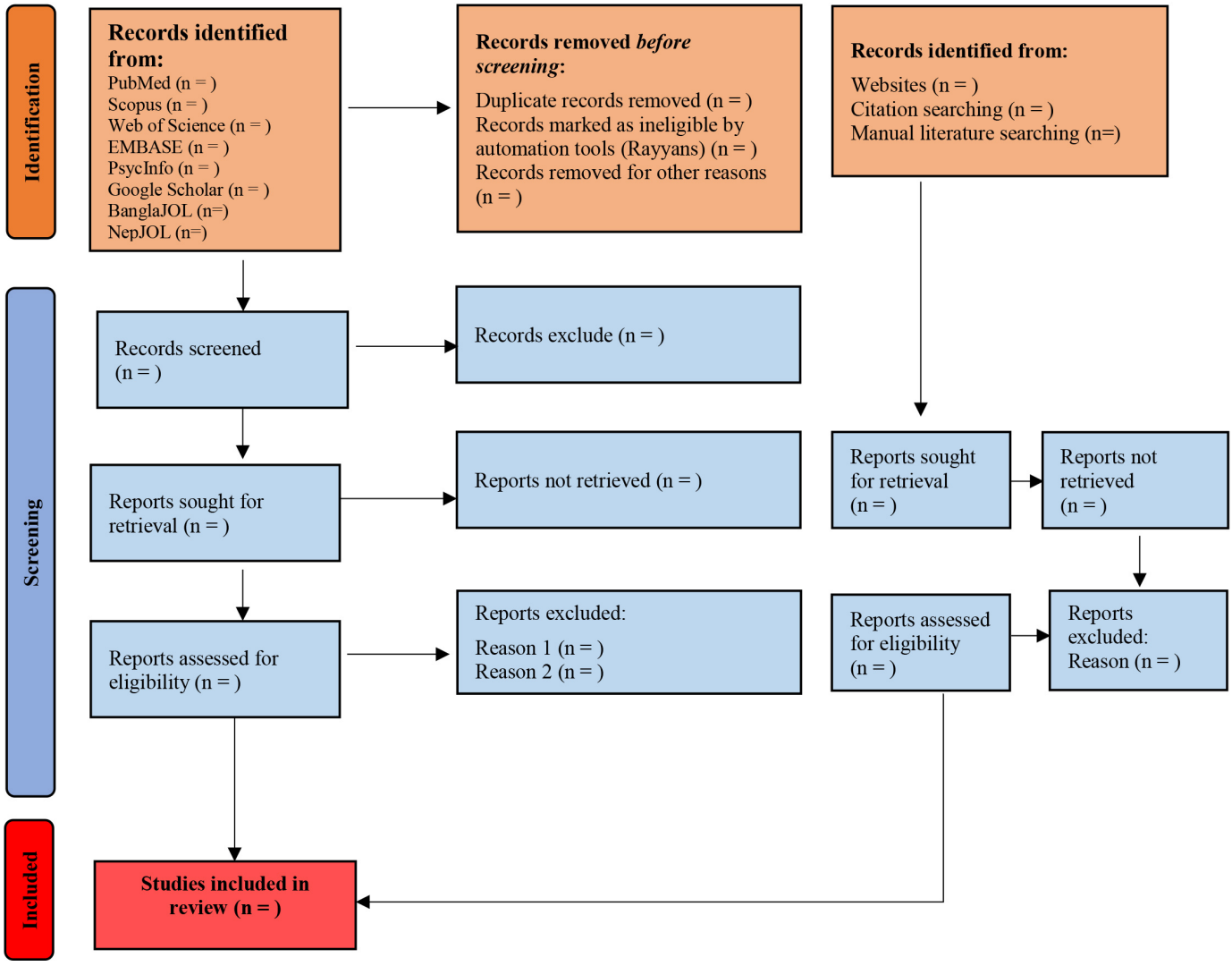


Figure 2 PRISMA 2020 flow diagram for systematic reviews including databases, registers and other sources. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

male vs female), country of origin, and type of institution (eg, public vs private). Additionally, studies using different validated assessment tools to assess suicidal behaviours (e.g., Columbia-Suicide Severity Rating Scale (C-SSRS), PHQ-9, BDI) will be grouped accordingly, considering each assessment tool as a separate subgroup to account for the potential influence of measurement tools on reported prevalence estimates.[14 22]

Missing/incomplete data

In instances where essential information is missing from the finally included studies, the review team will contact the respective corresponding authors via email to request clarification or additional data. If these efforts are unsuccessful, the incomplete data will be excluded from the quantitative synthesis (meta-analysis). However, any additional and available information from such studies will still be incorporated into the narrative synthesis section to ensure relevant findings are not overlooked. In cases where only a small portion of data is missing, and where appropriate, imputation technique will be employed to estimate the missing values. The choice will be based on the nature and extent of the missing data. All decisions regarding data imputation, exclusion or incorporation into the narrative synthesis, along with their potential impact on the overall findings, will be transparently addressed in the discussion section of the final review.

Sensitivity analysis

Sensitivity analyses will be conducted to assess the robustness of the pooled prevalence estimates by excluding studies rated as low quality during the quality assessment process.[40] This will be particularly applied to the meta-analyses of lifetime prevalence of SI, SA and SP to determine whether study quality significantly influences the overall findings. The meta-analyses will be re-run after the exclusion of low-quality studies, and changes in the pooled estimates and confidence intervals will be evaluated. Differences in pooled effect sizes will be compared using random-effects models. If notable differences are observed, it may indicate the influence of study quality on the overall study findings.

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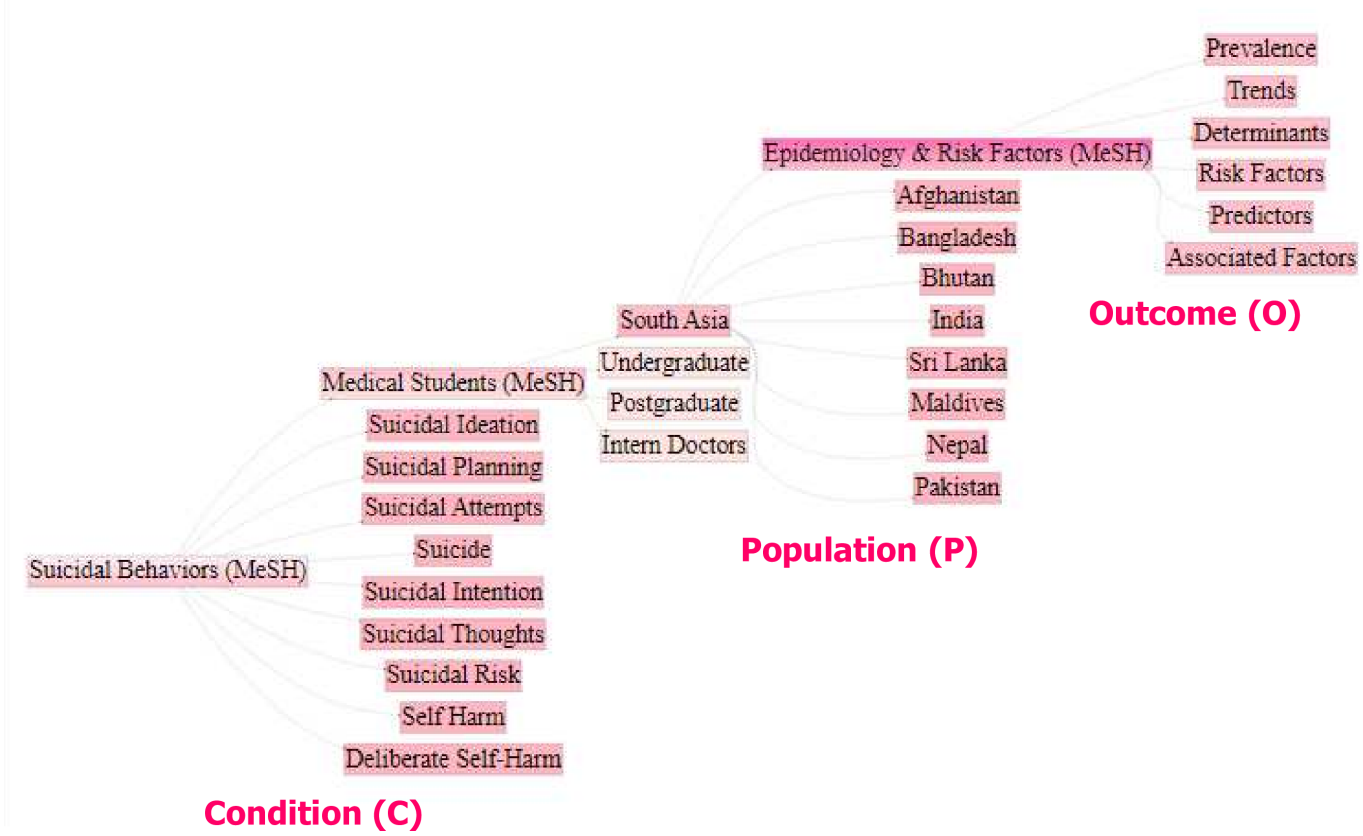


Figure 3 PICO (**Population, Intervention, Condition, Outcome**). framework and tree network plot of key search terms (MeSH terms).

Risk of bias (quality) assessment of individual studies

The methodological quality and certainty of the overall quality of the review will be assessed using the Grading of Recommendations, Assessment, Development and Evaluation approach, which is suitable for evaluating evidence on risk factors in this systematic review.[41] Additionally, a modified version of the Newcastle-Ottawa scale (mNOS),[42] having the scores allocated for each section with a total score of 10, will be applied to evaluate methodological quality across different types of observational studies. The scores for each section will be summed up and studies will be categorised as high risk of bias (≤ 3), moderate risk of bias (4 – 6), and low risk of bias (≥ 7), while high, moderate and low risk of bias studies will be considered for the final inclusion.[42] Key quality assessment criteria will include three domains: (1) sample

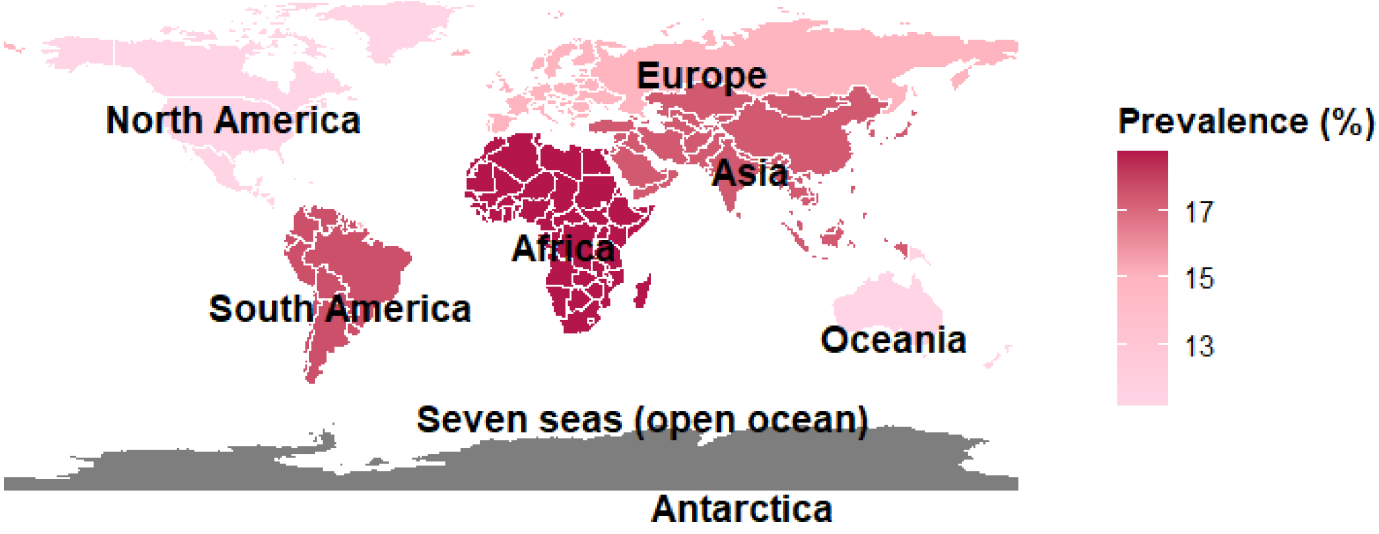


Figure 4 Prevalence of suicidal behaviours among medical students across the global regions The world map showing the prevalence of suicidal behaviours among medical students across regions was created using R v.4.3.2 and is based on data extracted from the aforementioned published studies. Data sources: Rotenstein *et al*;[1] Dyrbye *et al*;[2] Rahman *et al*.[5]

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representativeness, sample size, (2) comparability of the different outcome groups and (3) outcome assessment and the appropriateness of statistical analysis. Two independent reviewers will conduct the risk of bias assessment. Discrepancies will be resolved through discussion or by consulting a third reviewer if consensus is not reached.

Strategy for data synthesis

To investigate the prevalence of suicidal behaviours among medical students in South Asia, a comprehensive electronic search will be conducted across six major databases. A preliminary search in PubMed on 12 April 2025, retrieved 20 potential studies and co-occurrence of key search terms, outcome, explode functions applied to optimise the search process are visualised in a tree network from in figure 3. Following screening and eligibility assessment, data from relevant studies will be systematically extracted and organised for quantitative synthesis. In addition, demographic characteristics (eg, age, sex, academic year, academic education including MBBS/BDS, institution, year of data collection), suicidal behaviours, assessment tools, quality assessment, methods used for suicidal behaviours and possible reasons or factors related to the suicidal behaviours will also be extracted during data extraction process. All the extracted data on demographics and others will be tabulated in the main result. To address heterogeneity across the studies, subgroup analysis on gender, secular trend, academic year, academic curriculum (MBBS/BDS) will be conducted based on the lifetime prevalence of SAs, SI and suicidal plan. If quantitative data on reason and risk factors of suicidal behaviour are extracted from the included studies, further meta-analysis of risk estimates and subgroup analysis (by region, type of suicidal behaviours, academic year, curriculum (MBBS/BDS) and gender) will be done. If quantitative synthesis is not feasible, a narrative synthesis will be performed to identify factors associated with suicidal behaviours among South Asian medical students.

Publication bias assessment

Rank correlation test (Begg's test),[43] Egger's linear regression methods for funnel plot asymmetry[44] and visual inspection of funnel plot will be used to assess the publication bias. And if the number of the available studies per primary outcomes is at least 10 studies, a funnel plot will be performed. Furthermore, a 'trim and fill' procedure will be carried out to assess the possible impact of publication bias using the Egger's test. The findings from Egger's test indicating significant asymmetry in the funnel plot, suggesting the presence of publication bias. A non-significant p-value (> 0.05) in both the rank-correlation and Egger's tests will be interpreted as indicating low risk of publication bias.[43 44] The assessment of bias will be evaluated and visualised via the R statistical software using the 'metafor' package.[45]

Patients and public involvement

This systematic review and meta-analysis protocol does not directly involve individual patients or public organisation.

The status and timeline of the study

The systematic review is currently in its early phase, focusing on refining the research question, developing the search strategy and identifying relevant databases and potential literature. The authors will identify the screening and data extraction process and anticipate that the full review, including synthesis and reporting of findings, will be completed within the next 15–18 months. This timeline allows for a thorough and rigorous evaluation of the available evidence, quality assessment and preparation of the manuscript for publication.[46]

Discussion and implication

Globally, the prevalence of SI among medical students is very high ranging from 3.8% to 18.7%,[5] with studies indicating a substantial burden in LMICs compared with general population and university graduates (figure 4). In addition, limited research, scarcity of surveillance system, lack of national suicidal database and under-reporting in many countries were identified until to assess the suicidal behaviours across the region.[5 13] Gender differences are also notable in many studies, with female students often reporting higher levels of psychological distress even twice compared with male.[5] By assessing whether South Asian data align with or diverge from global patterns, this review may help uncover region-specific risk profiles. If new trends emerge, the review will investigate underlying sources of heterogeneity. By drawing from a broad base of data across South Asia countries, this study aims to provide a comprehensive understanding of suicidal thoughts and behaviours (STB).[13 14 47–49] In addition, it will also explore the transition across STB spectrum and find the prevalence of SI, SP and SA further. This STB transition was also evident in recent meta-analysis among university students.[3] Furthermore, the evidence will guide academic institutions in implementing proactive interventions to address suicidal behaviours, ultimately contributing to improved student well-being and the prevention of suicide among future healthcare professionals.

Strengths and limitations of the review

This will be the first systematic review and meta-analysis to evaluate and synthesise evidence from longitudinal and observational studies on the prevalence of suicidal behaviours among medical students of South Asia. This study will follow established guidelines including the Cochrane Handbook,[50] PRISMA-P,[24] MOOSE[25] and mNOS[42] guidelines, to ensure rigorous screening, reporting and quality assessment. Study selection, data extraction and quality evaluation will be performed independently by two reviewers in a blinded manner, with oversight by a third reviewer to maintain impartiality.

To mention a few potential limitations of this study, first, the expected heterogeneity among existing studies

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distinct methods, which may challenge the meta-analytical process and affect generalising the study findings. Second, sample size, sampling frame and instruments may affect the prevalence of suicidal behaviours from context to context. In addition, self-reported data may introduce methodological biases. Lastly, based on the preliminary searching among the South Asian countries, few countries still lack adequate evidence to report regarding medical students' suicidal behaviours which could be another potential limitation of the study. However, to enhance the robustness and generalisability of the findings, missing data will be addressed through appropriate imputation techniques, or by excluding studies with substantial gaps.

ETHICS AND DISSEMINATION

This review protocol is based solely on a synthesis of previously published research, no primary data collection will be conducted, and therefore, ethical approval is not required. The review will be conducted in accordance with the following PRISMA-P and PRISMA reporting guidelines, and the findings will be compiled into a manuscript for submission to an indexed international peer-reviewed journal. In addition to academic publication, the key results of the review will be shared through national and international scientific conferences, webinars and other relevant forums. To maximise the impact of this study, key findings will also be disseminated to policymakers, mental health professionals, academic institutions and other stakeholders to inform concerned medical education authority, clinical practice, guide future research and support the development of effective suicide prevention strategies.

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